

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date: September 28, 2026	Team ID: SWTID-2026-9692
Project Name: AI WeatherWise API	Marks: 4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	- Registration via JSON API endpoint (<code>/api/auth/register</code>) with name, email (<code>shree@gmail.com</code>), and password (<code>1234567</code>). - Creation and storage of unique MongoDB User Object ID (<code>_id</code>). - Automatic password hashing prior to database persistence.
FR-2	User Authentication & Security	- User Login validation via endpoint (<code>/api/auth/login</code>). - JSON Web Token (JWT) Bearer Token generation upon successful authentication. - Middleware route guarding for protected API endpoints using JWT authentication headers.
FR-3	Weather Data Services	- Retrieve city weather metrics via GET endpoint (<code>/api/weather/chennai</code>). - Return structured weather attributes including temperature (32.8°C), humidity (63%), wind speed (8.41), and condition ("Clouds").
FR-4	AI Weather Engine	- Generate natural language weather summaries via POST endpoint (<code>/api/ai/weather-summary</code>). - Generate tailored clothing and activity recommendations via POST endpoint (<code>/api/ai/weather-recommendation</code>). - Integrate with Google Gemini API (<code>gemini-3.8-flash</code>) for real-time model inferences.

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clear, standardized RESTful JSON API response bodies with explicit success flags and HTTP status codes (200 OK, 201 Created).

NFR No.	Non-Functional Requirement	Description
NFR-2	Security	Hashed password storage in MongoDB, stateless JWT authentication token verification, parameter sanitization against NoSQL injection, and CORS headers.
NFR-3	Reliability	Graceful error handling and fallback mechanisms for external API dependencies (such as Google Gemini API calls) to avoid server crashes.
NFR-4	Performance	Fast response times for API requests (sub-200ms execution for local Auth endpoints and sub-2.5s execution for AI inference requests).
NFR-5	Availability	Continuous API service operation on port 5001 with active MongoDB connectivity (127.0.0.1) and updated model endpoint paths.
NFR-6	Scalability	Stateless Express server architecture capable of scaling horizontally and delegating database persistence directly to MongoDB instance clusters.